

Department:	Computer Science
Course Lecturers:	Dr. Desmond Moru & Mr Chukwudi Ofoma
Course Code:	COS 101
Course Title:	Introduction to Computer Science
Course Unit:	3 Units: LH 30, PH 45
Semester:	First Semester
Time:	
Location:	Computer Lab I

Course Objective

The objectives of the course are:

- The aim is to help the students grasp the meaning of computer information systems in the digital age, the role computer plays in binary processing systems, how data is transmitted efficiently, computer programming, database management, computer networking, Internet of things, artificial intelligence, security and policy matters concerning information management.
- To understand the historical perspective of computing, hardware, software, information systems, and human resources and explore their integration and application in business and other segments of society.
- Complete lab assignments using the PC's operating system, and several commonly used applications, such as rust programming, databases, word processing and other applications.
- Complete lab practices on computer programming concepts.

Course Learning Outcomes

On completion of the course the students should be able to:

- Distinguish between computer and computer programming,
- Develop some techniques in computer science,

- Understand the different areas of study in computer science,
- Conversant with applications of computer science,
- Navigate the career prospects in computer science.
- Conversant with computer programming concepts.

Method of Lecture Delivery/Teaching Aids

- Lectures
- Practice and Programming Session
- Assignment
- Quizzes and Test

Structure of Programme/Method of Grading

Method of Assessment	Weighting %	Outline Details
Continuous Assessment Tests	30	Project presentation, Assignments
End of Semester Examination	65	Formal Examination
Class Participation	5	GitHub Commit

Course Outline

Week	Topic(s)	Activity
1	Historical perspective of Computer Science Installation Packages	• Lecture • Practice
2	Computer Algorithm Introduction to Rust – Environment Setup	• Lecture • Practice
3	Operating Systems Rust Cargo & Data Type	• Lecture • Practice
4	Computer Networking/Shell Line Commands Rust Decision Making & Loops	• Lecture • Practice/Project

5	Continuous Assessment (CA) 1	
6	Artificial Intelligence & Robotics Rust – Strings & Operators	• Lecture • Practice/Project
7	Internet of Things (IoT) Rust – Functions & Arrays	• Lecture • Practice/Project
8	Rust Vectors & Tuples	• Practice/Project
9	File Input & Output	• Practice/Project
10	Ownership & Structure	• Practice/Project
11	Database Management System/CA2	• Lecture
12	Introduction to PostgreSQL	• Lecture • Practice/Project
13	PostgreSQL & Rust	• Lecture • Practice/Project
14	Responsible Use of AI	• Lecture

Textbooks

Required Texts:

- *Computer Science: An Overview by J. Glenn Bookshear and Dennis Brylow*
- *Understanding Computers - Today and Tomorrow: by Deborah Morley and Charles Parker*
- *The Rust Programming Language: by Steve Klabnik and Carol Nichols*
- *Practical Rust Projects Building Game, Physical Computing, and Machine Learning Applications by Shing LYu*
- *Learning PostgreSQL Create, develop and manage relational databases in real world applications using PostgreSQL: by Salahaldin Juba, Achim Vannahme, and Andrey Volkov*